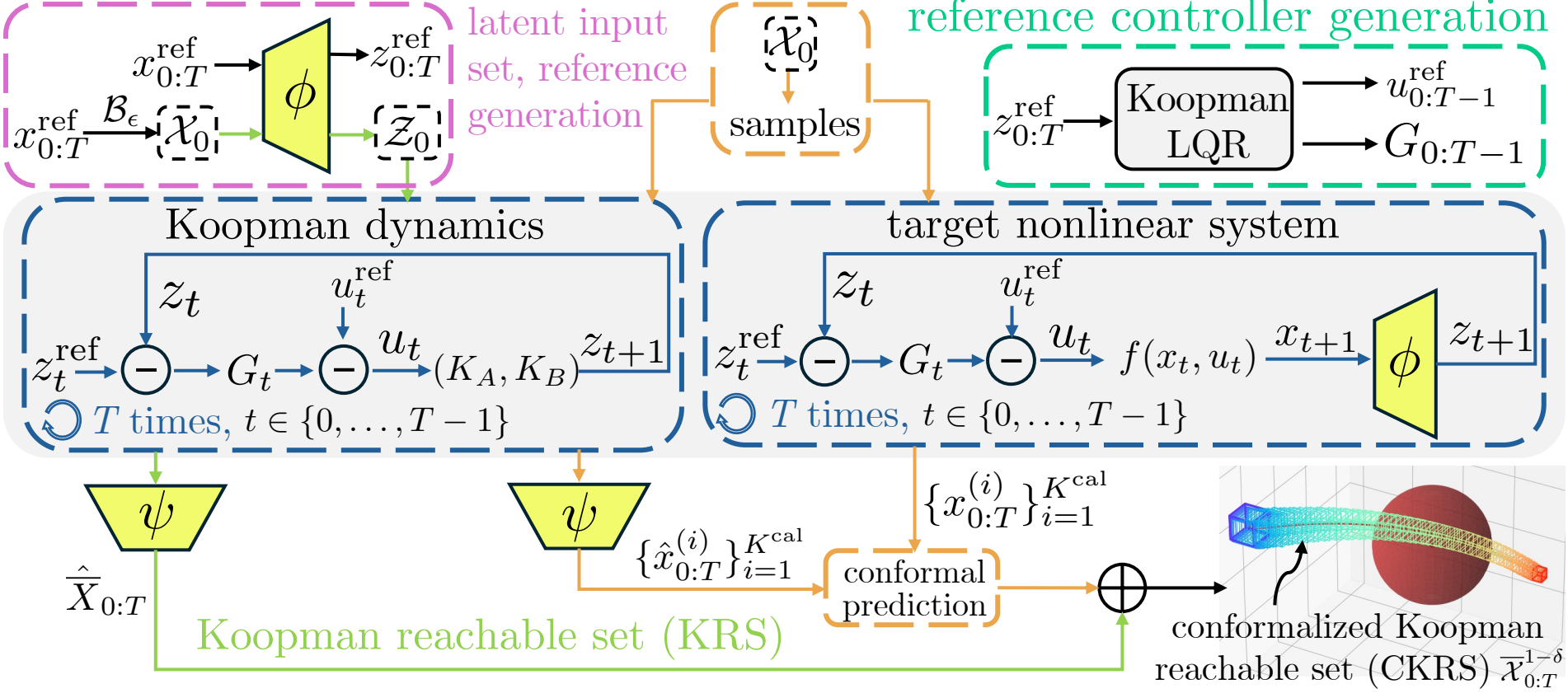
latent input set, reference generation
$x^{\text{ref}}_{0:T}$
$\mathcal{B}_\epsilon$
$\hat{\mathcal{X}}_0$
ϕ
$z^{\text{ref}}_{0:T}$
$\hat{\mathcal{Z}}_0$
$\hat{\mathcal{X}}_0$
samples
reference controller generation
$z^{\text{ref}}_{0:T}$
Koopman LQR
$u^{\text{ref}}_{0:T-1}$
$G_{0:T-1}$
Koopman dynamics
z_t
u^{ref}_t
z^{ref}_t
G_t
u_t
(K_A, K_B)
z_{t+1}
T times, $t \in \{0, \ldots, T-1\}$
target nonlinear system
z_t
u^{ref}_t
z^{ref}_t
G_t
u_t
$f(x_t, u_t)$
x_{t+1}
ϕ
z_{t+1}
T times, $t \in \{0, \ldots, T-1\}$
ψ
ψ
$\{x^{(i)}_{0:T}\}_{i=1}^{K^{\text{cal}}}$
$\{\hat{x}^{(i)}_{0:T}\}_{i=1}^{K^{\text{cal}}}$
conformal prediction
$\hat{\overline{X}}_{0:T}$
Koopman reachable set (KRS)
conformalized Koopman reachable set (CKRS) $\overline{\mathcal{X}}^{1-\delta}_{0:T}$